Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology (B.Sc.I.T Semester III)

Python For Data Science Practical

```
Practical - VI
```

Roll No.: S021

Name:LODHI IQRA

Class:SYBsc IT

Subject: PYTHON FOR DATA SCIENCE

Date of Assignment:19/08/2026

Date/Time of Submission:19/08/2026

bold text

PART – B Practical Questions:

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records. **bold text**

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

```
print(df.head())
```

```
   Roll No     Name Department  Marks  Attendance
0      101  DIVYESH        BCA   78.0        99.0
1      102   DHARMI      BTECH   80.0        90.0
2      103   HEMISH        BCA   88.0        92.0
3      104     SMIT     BSc CS   68.0        88.0
4      105   URVISH     BSc IT   45.0        70.0
```

```
print(df.tail())
```

```
   Roll No    Name Department  Marks  Attendance
1      102  DHARMI      BTECH   80.0        90.0
2      103  HEMISH        BCA   88.0        92.0
3      104    SMIT     BSc CS   68.0        88.0
4      105  URVISH     BSc IT   45.0        70.0
5      106     NaN        NaN    NaN         NaN
```

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information. **bold text**

```
print(df.shape) #number of rows and columns
```

```
(6, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```
Roll No         int64
Name           object
Department     object
Marks         float64
Attendance    float64
dtype: object
```

```
print(df.describe)
```

```
<bound method NDFrame.describe of    Roll No     Name Department  Marks  Attenda
0      101  DIVYESH        BCA   78.0        99.0
1      102   DHARMI      BTECH   80.0        90.0
2      103   HEMISH        BCA   88.0        92.0
3      104     SMIT     BSc CS   68.0        88.0
4      105   URVISH     BSc IT   45.0        70.0
5      106      NaN        NaN    NaN         NaN>
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000. **bold text**

```
result = df[df["Attendance"]>=60]
print(result)
```

```
   Roll No     Name Department  Marks  Attendance
0      101  DIVYESH        BCA   78.0        99.0
1      102   DHARMI      BTECH   80.0        90.0
2      103   HEMISH        BCA   88.0        92.0
3      104     SMIT     BSc CS   68.0        88.0
4      105   URVISH     BSc IT   45.0        70.0
```

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price. **bold text**

```
result = df.sort_values("Marks")
print(result)
```

```
   Roll No     Name Department  Marks  Attendance
4      105   URVISH     BSc IT   45.0        70.0
3      104     SMIT     BSc CS   68.0        88.0
0      101  DIVYESH        BCA   78.0        99.0
1      102   DHARMI      BTECH   80.0        90.0
2      103   HEMISH        BCA   88.0        92.0
5      106      NaN        NaN    NaN         NaN
```

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column. **bold text**

```
print("Maximum Marks ",df["Marks"].max())
print("Minimum Marks ",df["Marks"].min())
```

```
Maximum Marks  88.0
Minimum Marks  45.0
```